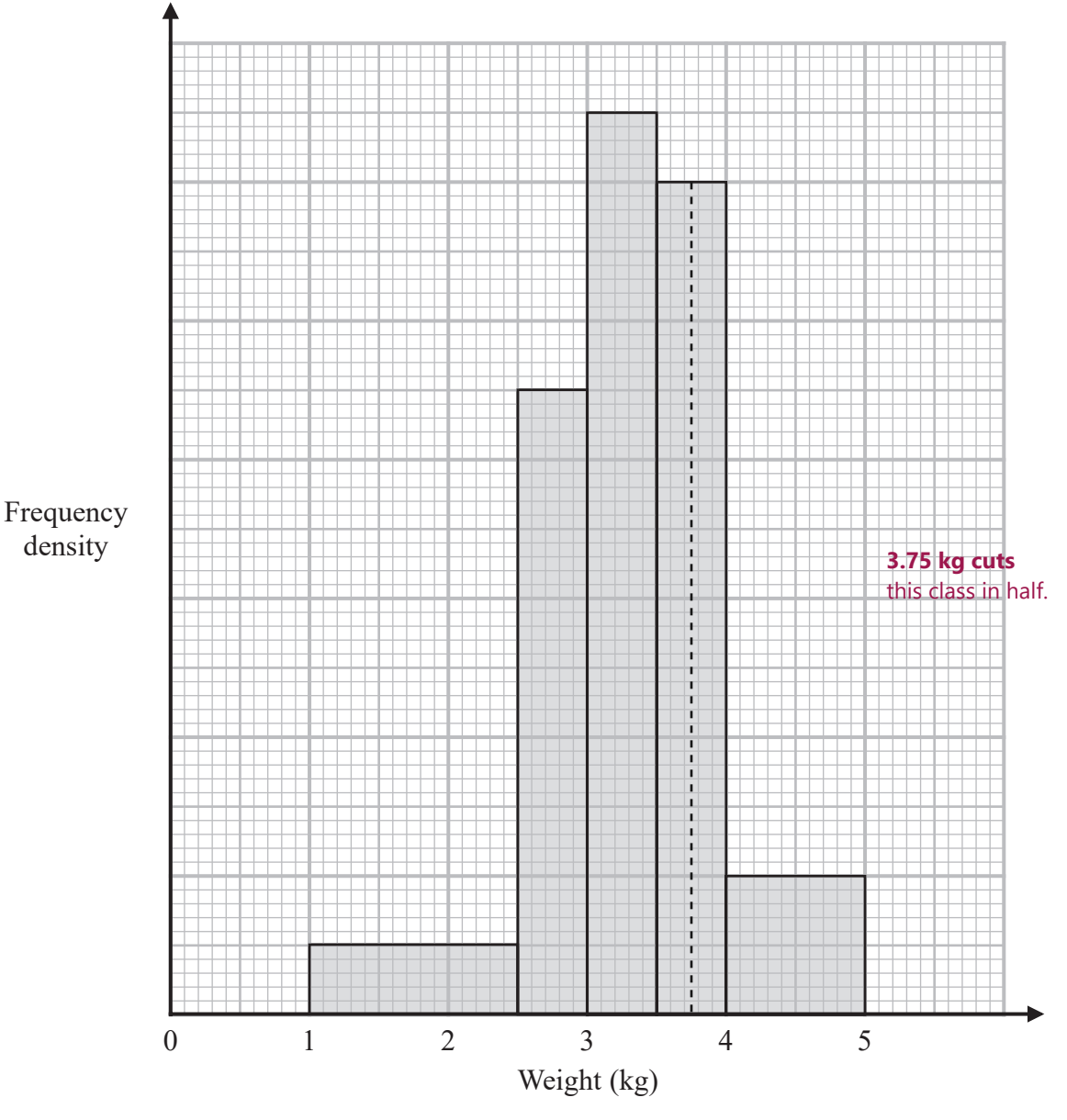
19 The histogram shows information about the weights, at birth, of some babies born in a hospital.



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12 of the babies weighed less than 2.5 kg at birth.

(i) Work out an estimate for the number of babies that weighed more than 3.75 kg at birth.

Below 2.5 kg: area 3 cm^2 represents 12 babies.

So 1 cm^2 represents 12 ÷ 3 = 4 babies.

1 mark
Frequency follows BAR AREA.

Above 3.75 kg: area = ½ × 12 + 4 = 10 cm^2

1 mark
Half of 3.5–4 kg plus all of 4–5 kg.

Estimated number = 10 × 4 = 40 babies

1 mark
Use the same area-to-frequency scale.

40

(3)

(ii) Explain why your answer to part (i) is only an estimate.

The calculation assumes babies are evenly spread within the 3.5–4 kg class. The histogram does not tell us exactly how many are above 3.75 kg within that class.

(1)

1 mark
The class must be split, so the result is an estimate.

(Total for Question 19 is 4 marks)

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